Name: Dallas, Eliana, and Michael **Date:** March 6th, 2025 **Class:** P1

MyDesign Portfolio

Element D Initial Template

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision-making regarding which design to include, ideally with a decision matrix.

Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document.

Generate a design blueprint with enough details that someone else not familiar with your project could also build your design. Think of this as your first sketch.

You will then proceed to build and test this design, returning to Element D when you are ready to generate more design ideas. Complete each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Underwater Adhesive

Name: Dallas, Eliana, and Michael **Date:** March 6th, 2025 **Class:** P1

Date: 3/6
Description:

- Some sort of waterproof tape where microplastics can be attached

Drawing that includes distinguishing information such as scale, dimensions, and materials:

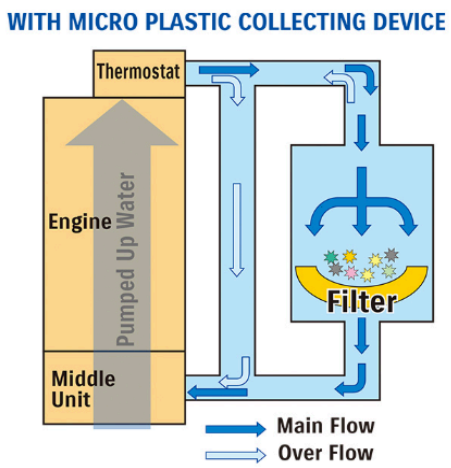
- Waterproof tape (duh)

Pump water through a filter

Date: 3/6
Description:

- Self-explanatory

Drawing that includes distinguishing information such as scale, dimensions, and materials:



- Image credit: https://www.globalsuzuki.com/suzuki_family/marine/suzuki_announced_the_development_of_micro-plastic_collecting_device.html
- 5L by 6W by 4.25H

Collapsable drawstring Container

Name: Dallas, Eliana, and Michael **Date:** March 6th, 2025 **Class:** P1

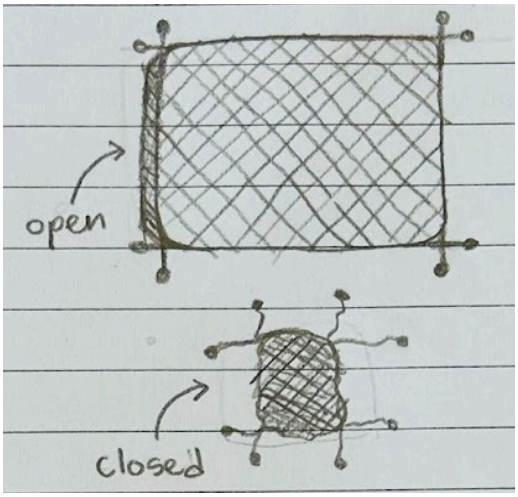
Date: 3/6

Description:

- Mesh container with drawstrings on all sides
- Keep container open when in water, drawstrings are pulled before container is taken out of the water
    - Holes become smaller when closed → keep microplastics in while water drains
- Connected to the ROV by a tether
- Pulled by the ROV, slightly behind the main body
- Open: 5.25 mm holes
- Closed: 0.5 mm holes

Drawing that includes distinguishing information such as scale, dimensions, and materials:

- 5L by 6W by 4.25H
- Mesh
- Braided cord
- Collapsible wall/corner supports



Name: Dallas, Eliana, and Michael **Date:** March 6th, 2025 **Class:** P1

Design Idea Comparison and Selection *(D1, D2)*

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify lothe weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

	Underwater Adhesive	Pump Water Through a Filter	Collapsible Drawstring Container
Criterion A (High)	3/5	3/5	2/5
Criterion B (High)	3/5	4/5	4/5
Criterion C (High)	4/5	3/5	2/5
Criterion D (High)	4/5	2/5	4/5
Criterion E (High)	4/5	2/5	3/5
Criterion F (Low)	2/3	2/3	2/3
Criterion G (Low)	3/3	1/3	2/3
Criterion H (Low)	3/3	2/3	3/3
Score	26/34	19/34	22/34

Name: Dallas, Eliana, and Michael **Date:** March 6th, 2025 **Class:** P1

Key

Criterion A (High Priority): Neutral buoyancy

Criterion B (High Priority): Collects microplastics

Criterion C (High Priority): Measure how fast water is moving in certain areas (sediment and water)

Criterion D (High Priority): Multiple ways of collecting microplastics

Criterion E (High Priority): Moves efficiently

Criterion F (Low Priority): Use as little plastic as possible

Criterion G (Low Priority): Low cost

Criterion H (Low Priority): Palatable size

High Priority: x/5

Low Priority: x/3

Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date: 3/11/2025
Sketch:
Manufacturing Methods:

Step 1: Make extension pipe off of ROV
Step 2: Wrap tape around the extension pipe
Step 3: Construct a box of mesh filters
Step 4: Attach the box around taped pipe

Supply List and Costs:

Name: Dallas, Eliana, and Michael **Date:** March 6th, 2025 **Class:** P1

Item	**Cost each**	**# needed**	**Total**
PVC Pipe	Free!	Roughly 14	Free!
Double-Sided Underwater Tape	$18.99	1	$18.99
1mm Mesh Filter	Free!	6 pieces	Free!
5 mm Mesh Filter	$5	2 pieces	$10
Total Cost			$29

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

After working on the Underwater Adhesive design, we realized that it will be extremely difficult to remove the microplastics from the tape. Hence, we have decided to change the microsplaystic collector designs. From now on, we will be making the Collapsible Drawstring Container.

Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date: 3/28/2025
Sketch: In D1 section of this document
Manufacturing Methods:

Name: Dallas, Eliana, and Michael **Date:** March 6th, 2025 **Class:** P1

Step 1: Make extension pipe off of ROV
Step 2: Wrap tape around the extension pipe
Step 3: Construct a box of mesh filters
Step 4: Attach the box around taped pipe

Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
PVC Pipe	Free!	Roughly 14	Free!
Double-Sided Underwater Tape	$18.99	1	$18.99
1mm Mesh Filter	Free!	6 pieces	Free!
5 mm Mesh Filter	$5	2 pieces	$10
Total Cost			$29

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.